

Beyond Prevalence: Mitigation Signals and Interventions for Histopathology Patch Classification and WSI-Level MIL

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Abstract

A classifier can trail on a class for several distinct reasons, and mitigation methods differ in which of them they assume: some infer the need for mitigation from class frequency, others from example difficulty. This report characterizes mitigation methods along two orthogonal axes: the signal that identifies a class or example as needing mitigation—prevalence, support, difficulty, or within-class diversity—and the level at which the intervention acts. Which methods apply depends on the input regime: labeled image patches support ordinary classification, whereas whole-slide image (WSI) diagnosis is commonly modeled as weakly supervised learning over patch-feature bags. Each method is therefore defined in the input form it consumes, with cross-entropy as the no-mitigation reference.

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1 Introduction

A pathology classifier can reach high aggregate accuracy while remaining unreliable on precisely the classes a diagnostic system is deployed to catch. When common tumor types dominate the training data, prevalence-sensitive evaluation rewards the classes that already dominate, so rare-class recall can degrade without the headline number moving. Long-tailed learning and medical-image evaluation literature therefore motivate classwise recall, macro F1, balanced accuracy, and calibration-aware probability metrics as primary outcomes in imbalanced settings (Saini and Susan, 2023; Zhang et al., 2023; Mosquera et al., 2024).

Established mitigation methods differ not only in the level at which they intervene but also in the signal they use to infer which observations require mitigation. Class frequency is one such signal, and a low class count conflates several distinct shortages: a small share relative to the other classes, a small absolute amount of support, intrinsic difficulty under the available representation, and narrow diversity of the class’s own feature manifold. Section 2 therefore makes both readings of the roster explicit, so that frequency is treated as a proxy for mitigation need rather than as its established cause. The methods themselves are drawn from the class-imbalance and long-tailed learning literature and are defined here in the order of their intervention level. The aim is to fix the method definitions, assumptions, and fidelity qualifications used when instantiating these methods on specific pathology datasets.

Prediction regimes. Which methods apply depends on the input regime, so each method is defined in the input form it consumes. In the *patch* regime, each labeled crop is a training and prediction unit—not an independent observation, since crops from one case share its acquisition and much of its morphology (Section 2.3): a frozen foundation-model encoder such as Virchow2 embeds the patch once, and a lightweight classifier head is trained on the resulting embedding (Zimmermann et al., 2024; Vorontsov et al., 2024). In the *WSI-bag* regime, a slide is represented as a bag of frozen patch-feature instances that share one weak slide-level label, and an attention-based MIL head aggregates the instances into a slide prediction, through attention, instance ranking, or bag-level representation learning. Both regimes consume embeddings from the same pathology foundation-model family but differ in supervision unit, and freezing the encoder isolates imbalance interventions from end-to-end backbone learning. The broader computational-pathology pipeline these regimes sit in—whole-slide imaging, patching into tiles, and weakly supervised multiple-instance learning over frozen foundation-model features—is introduced in the companion datasets report (Queisler, 2026).

2 Taxonomy

Two questions organize the tested methods. The first asks which signal a method consults to infer *that* a class or an example needs help; the second asks at which level of the training pipeline it acts on that inference. The questions are independent, and neither answers the other: class-weighted CE gives example i a weight $w_i = f(\pi_{y_i})$ read from *prevalence*, the share its class holds in the training set, while focal loss gives it a weight $w_i = f(p_{y_i})$ read from *difficulty*, the confidence the model already assigns to its label, so two methods that are both loss reweighting can rest on entirely different assumptions about what causes tail failure. This section fixes the notation used throughout and then defines both axes.

2.1 Notation

Let C be the number of target classes and let n_c denote the number of training examples from class c , so that $N = \sum_{j=1}^C n_j$ is the training-set size and $\pi_c = n_c/N$ is the empirical prevalence of class c . For a labeled patch or slide-level bag with label $y \in \{1, \dots, C\}$, the model outputs

logits $\mathbf{z} \in \mathbb{R}^C$ and class probabilities

$$p_c = \frac{\exp(z_c)}{\sum_{j=1}^C \exp(z_j)}. \quad (1)$$

For WSI-bag methods, a slide is represented as a bag $B = \{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ of frozen patch-feature instances. The attention-MIL backbone maps each bag to a bag embedding $\mathbf{b}(B)$ and class logits. Unless a method explicitly changes the sampler, objective, or representation loss, the empirical class frequencies in the training set define the optimization signal.

2.2 Intervention Levels

The intervention axis records where in the learning problem a method acts. Cross-entropy (CE) is the no-mitigation reference condition on this axis: it uses the observed training distribution as given and does not rebalance samples, labels, costs, or representations. The remaining methods group as data-level sampling, algorithm-level losses and prior corrections, hybrid sampling–loss methods, and representation or architecture-level methods (Saini and Susan, 2023; Zhang et al., 2023). Figure 1 places the tested roster in these families, and Section 3 defines the methods in that order, since a shared intervention level implies shared machinery to describe.

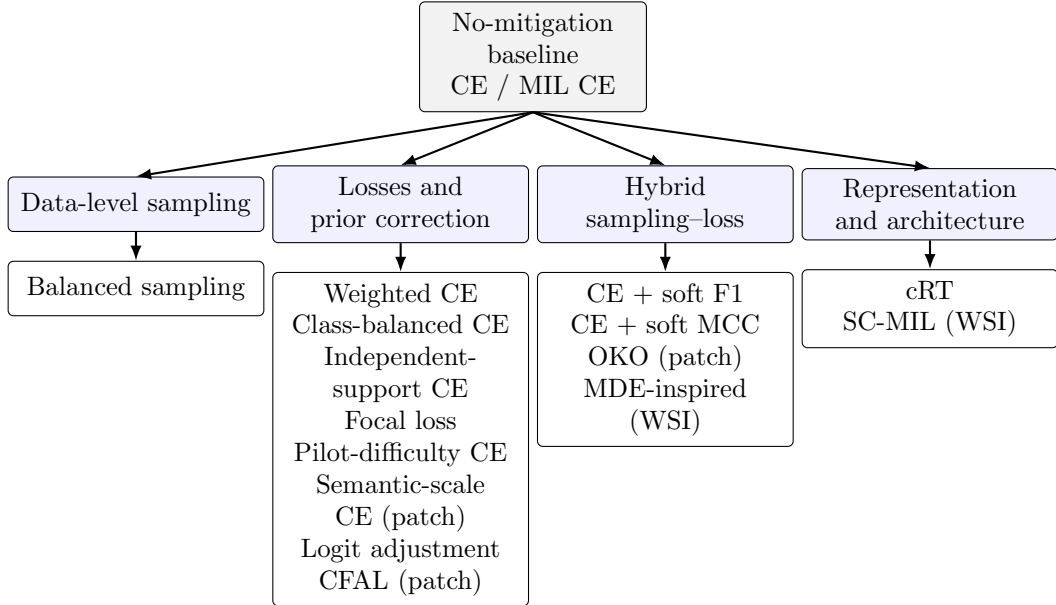


Figure 1: Intervention-level organization of the tested methods. Cross-entropy is the no-mitigation reference; alternatives intervene through sampling, objectives, prior correction, or representation and classifier design. Table 2 adds the orthogonal signal axis.

2.3 Mitigation Signals

The signal axis records what a method must observe before it concludes that mitigation is warranted, stated as the quantity the method actually consults rather than the property it ultimately affects.

Four signals are described in the following (Table 1): prevalence, support, difficulty, and diversity, with support splitting into a nominal and an independent reading. They are stated as properties of a class or example, independently of the intervention that acts on them.

Signal	What it describes	Representative quantity
Prevalence	Share of class c relative to the remaining classes	π_c , support rank, imbalance ratio ρ
Support	Absolute amount of material available for c , in either of two subtypes	<i>Nominal support</i> : n_c read absolutely rather than as a share, effective number $E_c(\beta)$. <i>Independent support</i> : distinct cases G_c or slides, with the redundancy-adjusted effective support $N_{\text{eff},c}$ as a descriptive refinement
Difficulty	How hard a class or example is to classify under a fixed representation	Loss, predicted confidence p_y , decision margin, class difficulty d_c
Diversity	How much within-class variation the observed examples of c exhibit <i>under a specified representation</i>	Feature-manifold volume S'_c (semantic scale), effective rank erank_c

Table 1: Signals a mitigation method can use to decide that a class or example needs help. Prevalence is relative and support is absolute, whether counted nominally or in independent units; difficulty and diversity are not determined by the class-size profile either, difficulty being read from model or probe behaviour and diversity from feature geometry, and both are read relative to a representation that has to be named with the estimate.

2.3.1 Prevalence

Prevalence is the share π_c of class c in the training distribution, together with the monotone summaries derived from it: the support rank of a class and the head-to-tail imbalance ratio $\rho = \max_c n_c / \min_c n_c$. It is a relative quantity. Rescaling every class count by a common factor leaves all prevalences unchanged, so a prevalence-aware rule behaves identically on a small and a large copy of the same class profile.

Prevalence is easily conflated with the amount of data a class actually carries, because both are usually read off the same counts n_c . The two come apart as soon as the profile changes shape rather than scale. Consider the training profiles $[8, 8, 8, 8]$ and $[64, 32, 16, 8]$. The scarcest class holds eight examples in both, so its nominal support is identical; its prevalence is $\frac{1}{4}$ in the first profile and $\frac{1}{15}$ in the second. A prevalence-aware rule treats that class very differently across the two profiles, whereas a rule driven by the absolute amount of material should treat it much the same. That absolute reading is the second signal.

2.3.2 Support

Support describes the absolute amount of training material available for class c , rather than its share of the dataset. Two readings are distinguished. *Nominal support* is the number of labelled examples n_c , or a saturating transform such as the effective number $E_c(\beta)$ of Equation (9). *Independent support* counts the distinct units from which those examples originate.

This distinction matters in pathology because observations are nested: cases contain slides, and slides contain patches. Here, a *case* denotes the patient-grouping unit over which dataset splits are drawn. Patches from the same case share acquisition conditions and morphology, while instances within a WSI bag also share one weak slide-level label. Consequently, many patches from a few cases provide less independent evidence than the raw patch count suggests. Independent support is therefore measured by the number of contributing cases, denoted G_c . At WSI level, slides are prediction units; cases remain independent units if one case contributes multiple slides.

For class c , let m_{gc} denote the number of examples contributed by case g , so that $n_c = \sum_{g=1}^{G_c} m_{gc}$. The case count G_c is model- and representation-independent but cannot express

within-case redundancy. This is described by the redundancy-adjusted effective support

$$N_{\text{eff},c} = \frac{n_c}{\text{DE}_c}, \quad \text{DE}_c = 1 + (m_c^* - 1) \text{ICC}_c, \quad m_c^* = \frac{\sum_{g=1}^{G_c} m_{gc}^2}{n_c}, \quad (2)$$

following the design-effect treatment of clustered observations (Kish, 1965; Rutterford et al., 2015). Here, m_c^* is the size-biased cluster size and ICC_c is the correlation between scalar scores of examples from the same case. Its one-way random-effects ANOVA estimator with the unequal-cluster-size correction is

$$\text{ICC}_c = \text{clip}_{[0,1]} \left(\frac{\text{MS}_{B,c} - \text{MS}_{W,c}}{\text{MS}_{B,c} + (\tilde{m}_c - 1)\text{MS}_{W,c}} \right), \quad \tilde{m}_c = \frac{n_c - \sum_{g=1}^{G_c} m_{gc}^2/n_c}{G_c - 1}, \quad (3)$$

where $\text{MS}_{B,c}$ and $\text{MS}_{W,c}$ are the between-case and within-case mean squares, and $\tilde{m}_c$ corrects for unequal case sizes (Kenny and La Voie, 1985). The estimate is undefined when fewer than two cases or no within-case degrees of freedom are available. When $\text{ICC}_c = 0$, all examples retain their nominal contribution; when $\text{ICC}_c = 1$, support is reduced to at most one effective observation per case.

To avoid mixing support with classification difficulty, the ICC is estimated from a classifier-independent score: the projection of each frozen Virchow2 embedding onto the leading principal direction of a PCA fitted jointly across classes on a fixed class-balanced reference. This scalarization and the exchangeable within-case correlation model are approximations, so $N_{\text{eff},c}$ is reported alongside G_c as a descriptive statistic and does not enter the training objective. Because a full embedding pass over every condition would repeat the same cost as the diversity draw at far larger scale, the ICC estimate is drawn from a capped sample of at most 200 cases and 200 patches per case per class, a deviation from Equation (3) as written that trades estimation precision for tractability on a purely descriptive statistic.

Class-balanced CE and CFAL read nominal support through $E_c(\beta)$, whereas independent-support weighted CE reads G_c through the same saturating transform (Section 3.3.3). This shared transform permits comparison of raw-example and independent-unit counts. Because every class weight is normalized to mean one, however, these methods can respond only to between-class support differences: equally supported profiles such as $[5, 5, 5]$ and $[50, 50, 50]$ both produce unit weights. Support-aware reweighting therefore does not address a dataset that is uniformly small.

2.3.3 Difficulty

Difficulty describes how hard an example or class is to distinguish under a specified representation or model, independently of how often it occurs. A frequent class can remain difficult because its morphology overlaps with another class, while a rare class can be easy when its features are distinctive. Difficulty is therefore not determined by prevalence or support.

At example level, difficulty can be read from the model during training through the loss, the predicted probability p_y of the true class, or the margin between the true-class score and its strongest competitor. These measures are dynamic: they change as the model learns and may also assign high difficulty to label noise or atypical observations.

At class level, difficulty is summarized by the error of a probe evaluated under support-controlled conditions. This report uses

$$d_c = 1 - \overline{\text{recall}}_c, \quad (4)$$

where $\overline{\text{recall}}_c$ is the classwise out-of-sample recall of a class-balanced probe fitted on frozen features. Holding representation and support fixed allows classes to be compared by separability rather than by frequency.

The resulting d_c is a representation-specific proxy rather than an intrinsic property of the diagnosis. It depends on the encoder, probe family, available material, and label quality. This fixed class-level reading also differs from condition-specific learnability, which changes when a training condition changes the evidence available for a class.

2.3.4 Diversity

Diversity describes how much within-class variation the observed examples exhibit under a specified representation. It differs from support, which counts available material, and from difficulty, which concerns how readily classes or examples can be distinguished. Two classes with equal support can therefore differ in diversity when one occupies a narrow feature region and the other spans several morphological patterns or preparation conditions.

The covariance spectrum of the class embeddings summarizes this variation. Let $\lambda_1 \geq \dots \geq \lambda_D$ be its eigenvalues and $\hat{\lambda}_i = \lambda_i / \sum_{j=1}^D \lambda_j$ their normalized values. The effective rank is

$$H(\hat{\lambda}) = - \sum_{i=1}^D \hat{\lambda}_i \log \hat{\lambda}_i, \quad \text{erank}_c = \exp(H(\hat{\lambda})), \quad (5)$$

This quantity measures how many directions contain substantial variation: it equals one when all variance lies along one direction and D when variance is distributed evenly across all directions (Roy and Vetterli, 2007). Terms with $\hat{\lambda}_i = 0$ contribute zero, and a class with identical embeddings is assigned $\text{erank}_c = 1$.

Effective rank measures directional dimensionality but ignores overall spread because normalizing the eigenvalues removes their common scale. The semantic-scale volume S'_c instead uses a log-determinant and responds to both the number of occupied directions and their magnitude (Ma et al., 2023). Two classes can consequently share an effective rank while occupying very different feature-space volumes. The two statistics are complementary summaries of within-class geometry, not interchangeable definitions.

Diversity estimates depend on both representation and observed support. With too few examples, a covariance cannot reveal all occupied directions, so comparisons at unequal support can mistake undersampling for low diversity. *Fixed diversity* is measured in a frozen embedding space, whereas *dynamic diversity* is measured in a representation that changes during training. Neither reading captures morphological modes absent from the observed material. Diversity is therefore a representation- and sample-specific description of observed within-class variation, not a complete property of the diagnosis.

2.3.5 How Methods Use the Signals

Table 2 is a compact map of the method roster. Each row answers two questions: what information tells the method where mitigation is needed, and what part of training does the method change in response? The first answer is the *mitigation signal*; the second is the *intervention*.

A signal must enter the sampler, objective, or decision rule to count. An effect produced by training is not itself a signal: SC-MIL may compact bag representations and CFAL may separate prototypes, but neither uses a diversity estimate to decide when or how strongly to act. Details of each mechanism follow in Section 3.

Method	Mitigation signal	Intervention
Both regimes <i>patch and WSI-bag</i>		
CE, MIL CE	None	None (reference)
Balanced sampling	Prevalence (n_{y_i})	Sampling probabilities
Weighted CE	Prevalence (n_c)	Loss weight
Logit adjustment	Prevalence ($\hat{\pi}_c$)	Logits and prior correction
cRT	Prevalence (balanced exposure)	Classifier retraining
Class-balanced CE	Nominal support ($E_c(\beta)$)	Loss weight
Focal loss	Nominal support (n_c) and example difficulty (p_y)	Loss weight
Pilot-difficulty weighted CE	Class difficulty (d_c)	Loss weight
Patch regime <i>only</i>		
Independent-support weighted CE	Independent support (G_c)	Loss weight
Semantic-scale weighted CE	Dynamic diversity (S'_c)	Loss weight
CFAL	Nominal support and example difficulty	Prototype head and margin loss
CE + soft F1	Prevalence (balanced sampling)	Sampler and auxiliary loss
CE + soft MCC	Prevalence (balanced sampling)	Sampler and auxiliary loss
OKO	Prevalence (uniform pair class)	Set construction and set loss
WSI-bag regime <i>only</i>		
SC-MIL	Label structure and prevalence	Contrastive loss and CE curriculum
MDE-inspired	Prevalence (balanced exposure)	Dual experts and consistency

Table 2: Mitigation signal and intervention of each tested method, grouped by prediction regime.

Two patterns organize the table. First, prevalence is the most common signal, but prevalence-aware methods act through different mechanisms: sampling, loss weights, logits, classifier re-training, set construction, or multiple experts. Sharing a signal therefore does not make two methods mechanistically equivalent.

Second, five methods offer a more direct comparison of signals because they all apply a class weight to otherwise unchanged CE. Weighted CE reads prevalence, class-balanced CE reads nominal support, independent-support weighted CE reads case support, pilot-difficulty weighted CE reads class difficulty, and semantic-scale weighted CE reads diversity. Class-balanced and independent-support weighting isolate the counting unit further by sharing the same effective-number transform. Semantic-scale weighting uses the same class-weighting intervention, but unlike the other methods, it updates its signal as the learned representation changes.

3 Methods

Each method is defined below in the input form it consumes, following the intervention families of Figure 1. The comparison is regime-specific. Patch methods consume frozen Virchow2 embeddings of controlled patches and optimize one shared MLP classifier head; CE-based patch methods use the same linear output layer, while CFAL replaces that layer with class prototypes and affinity-based prediction. WSI-bag methods consume complete eligible frozen-feature bags and optimize one shared attention-MIL head. Both regimes therefore rely on the same pathology foundation-model family, but they differ in supervision unit: one embedding per selected patch versus aggregation over every eligible instance on a slide, without bag truncation.

3.1 Cross-Entropy Baseline

Unweighted cross-entropy is the baseline because it is the standard supervised objective without an explicit class-imbalance correction. For an example with label y , the loss is

$$\mathcal{L}_{\text{CE}}(\mathbf{z}, y) = -\log p_y. \quad (6)$$

In the patch regime, this objective trains the shared linear head on frozen Virchow2 patch embeddings from the controlled training set. In the WSI-bag regime, the same objective trains the shared attention-MIL model on slide-level feature bags. In both cases, CE uses the observed class distribution directly: majority classes appear more often in minibatches and therefore contribute more loss terms over an epoch. This makes CE an appropriate reference for asking whether an imbalance-specific intervention actually improves the rare-class tradeoff.

3.2 Data-Level Sampling

Data-level methods change what the optimizer sees, rather than changing the mathematical penalty assigned to one observed example. In the taxonomy, this family includes the resampling strategies that make minority examples more visible during training (Saini and Susan, 2023).

3.2.1 Balanced Sampling

Balanced sampling changes minibatch composition by sampling training examples with probabilities related to their class support through a strength $s \in [0, 1]$. For an example i with class y_i , the sampler weight is proportional to

$$q_i(s) \propto n_{y_i}^{-s}. \quad (7)$$

After normalization across the training set, $s = 0$ recovers ordinary sampling and $s = 1$ gives equal expected exposure to every class; intermediate values partially rebalance the data. The

patch variant applies this sampler to patch examples and still optimizes CE. The MIL variant applies the same idea at slide-bag level and also optimizes CE. Balanced sampling therefore tests a pure data-level intervention without changing the loss formula.

3.3 Algorithm-Level Losses

Algorithm-level methods keep the training examples fixed but alter how errors are penalized. They are cost-sensitive or difficulty-sensitive objectives: minority examples may receive larger explicit weights, or easy examples may contribute less to the gradient (Saini and Susan, 2023; Scholz et al., 2024).

3.3.1 Class-Weighted Cross-Entropy

Class-weighted CE multiplies each example’s CE term by a class-dependent cost controlled by the same strength $s \in [0, 1]$:

$$\mathcal{L}_{\text{WCE}}(\mathbf{z}, y) = -w_y \log p_y, \quad w_c(s) \propto n_c^{-s}. \quad (8)$$

The implementation normalizes these weights so that their mean across classes is one. Here $s = 0$ recovers unweighted CE and $s = 1$ gives inverse-frequency weighting; intermediate values reduce the correction. This method leaves the sample order and model architecture unchanged, but increases the loss contribution of classes with fewer training examples. It is evaluated for both patch classification and attention-MIL.

3.3.2 Class-Balanced Cross-Entropy by Effective Number

Class-balanced (CB) cross-entropy replaces inverse-frequency weighting by a saturating notion of how much a class count is worth (Cui et al., 2019). Its premise is that additional examples of a class need not contribute linearly increasing information: as support grows, examples increasingly resemble ones already seen, so the marginal contribution of the next example diminishes. For class c with training count n_c , the effective number of samples is

$$E_c(\beta) = \frac{1 - \beta^{n_c}}{1 - \beta}, \quad 0 \leq \beta < 1, \quad (9)$$

and the class weight is proportional to its inverse. As for class-weighted CE, the implementation normalizes the weights to mean one across classes,

$$w_c^{\text{CB}}(\beta) = \frac{\tilde{w}_c^{\text{CB}}(\beta)}{\frac{1}{C} \sum_{j=1}^C \tilde{w}_j^{\text{CB}}(\beta)}, \quad \tilde{w}_c^{\text{CB}}(\beta) = \frac{1 - \beta}{1 - \beta^{n_c}}, \quad (10)$$

and optimizes

$$\mathcal{L}_{\text{CB}}(\mathbf{z}, y; \beta) = -w_y^{\text{CB}}(\beta) \log p_y. \quad (11)$$

The single control β sets the count scale at which those diminishing returns take effect. At $\beta = 0$ every effective number equals one and the objective reduces to ordinary CE; as $\beta \rightarrow 1$ the normalized weights approach inverse-count weighting. Intermediate values are the informative ones, because they separate a *saturating* count correction from the power-law correction n_c^{-s} of Equation (8): under the power law every doubling of class support changes the weight by the same factor, whereas under Equation (10) the weight stops responding once n_c exceeds roughly $(1 - \beta)^{-1}$. The reference default is $\beta = 0.999$; β is the imbalance control selected on validation.

Two properties make this method the cleanest available isolation of the effective-number mechanism. First, n_c enters absolutely rather than as a share, which makes CB-CE the nominal-support rule of the roster: unlike the prevalence-driven methods its weights change when a class profile is scaled rather than reshaped, and unlike independent-support weighted CE it asks only

how many examples a class holds, not how many units they came from. Here n_c is the prediction-unit count of the training condition: labeled patches for patch classification and labeled slides for WSI-level MIL. Second, CB-CE changes neither the sampler, the head geometry, nor the output semantics, while CFAL couples the same effective number to class prototypes, a margin objective, an affinity-based difficulty factor, and a diversity regularizer (Equation (28)). An effect of CFAL therefore cannot be attributed to its effective-number term alone, whereas an effect of CB-CE can. The method is evaluated in both regimes.

3.3.3 Independent-Support Weighted Cross-Entropy

Independent-support weighted cross-entropy (ISW-CE) tests whether class support is better measured by raw examples or by the independent cases that produced them. Class-balanced CE uses the example count n_c , whereas ISW-CE replaces it with the number of distinct cases G_c . This distinction matters for patch classification because many correlated patches may originate from only a few cases. ISW-CE is constructed for this thesis to represent the independent-support signal defined in Section 2.3.2.

The method otherwise retains the effective-number weighting of class-balanced CE:

$$E_c^{\text{ind}}(\beta) = \frac{1 - \beta^{G_c}}{1 - \beta}, \quad w_c^{\text{ISW}}(\beta) = \frac{1/E_c^{\text{ind}}(\beta)}{\frac{1}{C} \sum_{j=1}^C 1/E_j^{\text{ind}}(\beta)}, \quad \mathcal{L}_{\text{ISW}}(\mathbf{z}, y; \beta) = -w_y^{\text{ISW}}(\beta) \log p_y. \quad (12)$$

The weights therefore have mean one across classes. Setting $\beta = 0$ recovers ordinary CE, while $\beta \rightarrow 1$ approaches inverse-case weighting. Because case counts are generally much smaller than patch counts, β is selected on validation over a grid appropriate to the case-count scale.

Two comparisons with class-balanced CE serve different purposes. The validation-tuned comparison evaluates which rule performs better when each uses its selected β ; its result reflects both the counting unit and the selected saturation scale. The matched- β comparison holds the transform fixed, leaving the substitution of G_c for n_c as the only difference. Only this controlled comparison supports attribution to the counting unit itself.

ISW-CE is evaluated for patch classification. The redundancy-adjusted support $N_{\text{eff},c}$ from Equation (2) is reported descriptively but does not enter the objective. When every prediction unit belongs to a distinct case, $G_c = n_c$ and ISW-CE reduces to class-balanced CE; this applies to slide-per-case cohorts such as CAMELYON16 (Queisler, 2026).

3.3.4 Focal Loss

Focal loss reduces the contribution of confidently classified examples and concentrates optimization on examples that remain difficult. With focusing parameter γ and class weight α_y , the multiclass focal objective is

$$\mathcal{L}_{\text{focal}}(\mathbf{z}, y) = -\alpha_y (1 - p_y)^\gamma \log p_y. \quad (13)$$

The reference default is $\gamma = 2.0$ for both regimes; γ is the tuned control selected on validation while the rest of the training protocol is held fixed. α is fitted alpha-balanced: the derived, not tuned, mean-one inverse-frequency weight of Equation (8) at $s = 1$. The method is relevant to imbalance because many minority-class examples can remain hard under a classifier dominated by head-class gradients; combined with α -balancing it addresses both difficulty and class frequency, rather than difficulty alone.

3.3.5 Pilot-Difficulty Weighted Cross-Entropy

Class-difficulty based weighting (CDB-W) replaces class frequency by measured classwise prediction difficulty as the signal for cost-sensitive learning (Sinha et al., 2022). It separates the

assumption that a class is difficult because it is rare from the possibility that a comparatively frequent class remains intrinsically difficult under the available representation. In the source method, difficulty after epoch e is

$$d_{c,e} = 1 - A_{c,e}, \quad (14)$$

where $A_{c,e}$ is accuracy for class c on a separate balanced monitor set. The next epoch weights examples of that class by $d_{c,e}^{\tau_e}$, with τ_e scheduled from the imbalance in classwise performance. CDB-W is therefore dynamic: its class weights change as training changes which classes remain difficult. Unlike focal loss, which weights individual examples through p_y , CDB-W assigns one difficulty to each class. It can consequently prioritize a frequent but poorly separated class over a rare class that is already easy to recognize.

Thesis variant. A source-faithful implementation would evaluate a balanced monitor set after every epoch. The tested variant instead estimates the matched-support difficulty d_c of Equation (4) once from out-of-fold probe predictions and fixes it before constructing the training conditions.

With $\epsilon = 10^{-3}$ to prevent a zero weight for a perfectly separated class, the mean-one normalized weight is

$$w_c^{\text{PDW}}(\tau) = \frac{(d_c + \epsilon)^\tau}{\frac{1}{C} \sum_{j=1}^C (d_j + \epsilon)^\tau}, \quad (15)$$

and the objective is

$$\mathcal{L}_{\text{PDW}}(\mathbf{z}, y; \tau) = -w_y^{\text{PDW}}(\tau) \log p_y. \quad (16)$$

The exponent $\tau \geq 0$ is the single imbalance control: $\tau = 0$ recovers ordinary CE, while larger values emphasize classes that the probe found harder. The small ϵ affects only zero or near-zero difficulties. Because d_c remains fixed during training, the method is reported as *pilot-difficulty weighted CE (CDB-W-inspired)*, not as a reproduction of dynamic CDB-W. It cannot respond to changing class difficulty and inherits estimation error from the probe.

Fixing d_c also enables a controlled comparison with class-weighted CE. Under a difficulty-aligned assignment, both methods favor the same classes; under a difficulty-reversed assignment, they favor opposite classes. Their contrast therefore tests whether mitigation helps because a class is rare or because it is difficult. The sampler and architecture remain unchanged, and the method is evaluated in both regimes.

3.3.6 Semantic-Scale Weighted Cross-Entropy

Semantic-scale-balanced learning weights classes by the diversity of their learned features rather than by their frequency (Ma et al., 2023). A class with features spread across many directions has a larger *semantic scale* than one concentrated in a narrow region.

For class c , let $Z_c \in \mathbb{R}^{d \times M}$ contain M centred feature vectors from the current representation. Semantic scale is measured by the regularized log-volume

$$S'_c = \frac{1}{2} \log_2 \det \left(I + \frac{d}{M\epsilon^2} Z_c Z_c^\top \right). \quad (17)$$

The identity term keeps this measure finite when $M < d$, while ϵ sets the feature scale at which variation is resolved. Larger values indicate greater within-class spread in magnitude or direction.

The full source method also multiplies S'_c by a term derived from distances between class centres. That term is intentionally omitted because centre separation describes difficulty as well as diversity. The resulting within-class-only configuration is evaluated by Ma et al. (2023) as DSB-CE-1.

Classes with smaller volume receive larger mean-one weights. Let $r_c = S'_c{}^{-\tau}$ for a class with a valid volume estimate, and $r_c = 1$ (no reweighting) for a class without one:

$$w_c^{\text{SSB}}(\tau) = \frac{r_c}{\frac{1}{C} \sum_{j=1}^C r_j}, \quad \mathcal{L}_{\text{SSB}}(\mathbf{z}, y; \tau) = -w_y^{\text{SSB}}(\tau) \log p_y. \quad (18)$$

The exponent $\tau \geq 0$ is the only tuned control: $\tau = 0$ recovers ordinary CE and $\tau = 1$ gives inverse-volume weighting. Because the volumes are refreshed as the representation changes, this remains a dynamic weighting method.

Thesis variant. The method uses the 512-dimensional hidden representation of the shared MLP head and is reported as *semantic-scale weighted CE (DSB-CE-1-inspired)*. To isolate diversity from support, R rolling feature pools $Q^{(r)}$ are constructed from matched draws in which every class contributes the same number of cases and patches.

Training follows the source method’s three-stage timing with $n = 5$. A *reference pass* comprises $\lceil T/B \rceil$ optimizer updates. During the first pass, each pool is filled by traversing its matched draw once, while the loss retains unit class weights. During passes two through five, one minibatch in each pool is refreshed from the current hidden representation at every optimizer update, progressively replacing the early features, but class weights remain one. From the sixth pass onward, the pools continue to be refreshed at every optimizer update; after each refresh, their classwise volumes are recomputed and averaged:

$$S'_{c,t} = \frac{1}{R} \sum_{r=1}^R S'_{c,t,r}, \quad (19)$$

The resulting weights from Equation (18) are applied at that optimizer update. Since $U = 30\lceil T/B \rceil$, reweighting begins after $U/6$ and remains active for the final five sixths of the budget. Averaging volumes rather than covariances preserves the matched budget of each draw. If maintaining all R pools is too expensive, $R = 1$ is used and reported. Unlike the source’s recent-minibatch pool, the matched pools cannot recover class frequency through unequal class occupancy; this retained deviation prevents the diversity estimate from also encoding prevalence.

Feature scale is stabilized at each refresh before Equation (17) is evaluated. All matched features are divided by the pooled isotropic scale

$$s_t = \left(\frac{1}{d} \text{tr } \Sigma_t \right)^{1/2}, \quad \varepsilon = 1, \quad (20)$$

where Σ_t is the pooled covariance across classes and draws. A single scalar removes global magnitude drift without reweighting feature directions or erasing between-class differences. The precision is fixed, leaving τ as the only validation-selected control.

A draw with fewer than two usable features for a class is excluded. If no valid draw remains, or the averaged volume is zero, the previous estimate is retained; a class with no valid estimate at all – at the first refresh, or in every refresh since – keeps unit weight ($r_c = 1$ in Equation (18)) rather than a maximal fallback weight, since a degenerate draw carries no diversity evidence to reweight by. Such events are recorded. The method is evaluated only for patch classification because matched WSI-level class sets are too small for stable covariance estimates.

3.3.7 Logit Adjustment

Logit adjustment explicitly corrects for the empirical training prior (Menon et al., 2021). Let

$$\hat{\pi}_c = \frac{n_c}{\sum_{j=1}^C n_j} \quad (21)$$

denote the class prior estimated only from the current training condition. The control $\tau \geq 0$ determines the strength of the correction; $\tau = 0$ recovers ordinary CE. In the train-time form, adjusted logits are used inside the loss,

$$\mathcal{L}_{\text{LA-train}}(\mathbf{z}, y) = -\log \frac{\exp(z_y + \tau \log \hat{\pi}_y)}{\sum_{c=1}^C \exp(z_c + \tau \log \hat{\pi}_c)}. \quad (22)$$

The positive prior term makes head classes harder to fit during training, inducing a classifier whose unadjusted logits favor a class-balanced target. In the post-hoc form, a model is trained with ordinary CE and corrected only at inference,

$$p_c^{\text{LA}}(\mathbf{x}) = \frac{\exp(z_c(\mathbf{x}) - \tau \log \hat{\pi}_c)}{\sum_{j=1}^C \exp(z_j(\mathbf{x}) - \tau \log \hat{\pi}_j)}. \quad (23)$$

The sign difference follows from where the correction is applied: the train-time loss learns against prior-shifted logits, whereas the post-hoc rule removes the observed training prior from ordinary CE logits. Both patch and WSI regimes use the same definition at their respective prediction unit. Because either form deliberately changes the implied target prior, its probabilities should not be interpreted as calibrated for the natural test prevalence without an explicit target-prior assumption. Discrimination and calibration are therefore reported together, with post-hoc temperature scaling kept conceptually separate from prior correction.

Score variants with different interpretations are retained as separate outputs. For an ordinary CE score z_c^{CE} , post-hoc logit adjustment uses $z_c^{\text{CE}} - \tau \log \hat{\pi}_c$ for class-balanced decisions; this score is not reported as a probability calibrated to the natural target prevalence. Under the prior-shift assumption, probabilities for a target prior π_c^{target} instead normalize the logits

$$z_c^{\text{CE}} - \log \hat{\pi}_c + \log \pi_c^{\text{target}}. \quad (24)$$

For train-time adjustment, CE is applied to $g_c = z_c + \tau \log \hat{\pi}_c$. Because g_c approximates training-posterior logits, the corresponding target-prevalence logits are

$$z_c + (\tau - 1) \log \hat{\pi}_c + \log \pi_c^{\text{target}}. \quad (25)$$

The unadjusted z_c is the train-time method’s validation-selected discrimination score but is fully prior-stripped only when $\tau = 1$. The target prior is estimated separately on each natural-validation split and applied only to the corresponding test split. Temperature scaling is fitted on natural-validation NLL after the applicable target-prior correction and then applied unchanged to test logits. Raw, balanced-decision, target-prior-corrected, and temperature-scaled outputs remain separate; a single temperature corrects overall sharpness rather than class-specific prior mismatch.

3.3.8 Center-Focused Affinity Loss (CFAL)

Center-focused affinity loss (CFAL) replaces the linear softmax head with learnable class prototypes and an affinity-based training objective (Mahbub et al., 2024). The motivation is that minority histology classes can remain visually similar in embedding space even when a frozen encoder already separates them coarsely; pulling examples toward compact class centers and penalizing confusion with nearby wrong-class prototypes can therefore improve tail-class boundaries without changing the sampler.

The tested patch implementation keeps the shared MLP trunk that maps a frozen Virchow2 embedding $\mathbf{x}$ to a hidden representation $\mathbf{z} = f(\mathbf{x}) \in \mathbb{R}^{512}$, but replaces the final linear map with one learnable prototype $\mathbf{w}_c \in \mathbb{R}^{512}$ per class. Both embeddings and prototypes are L2-normalized before affinity computation. The Gaussian affinity between example i and class c is

$$a_{ic} = \exp\left(-\frac{\|\hat{\mathbf{z}}_i - \hat{\mathbf{w}}_c\|_2^2}{\sigma}\right), \quad \hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|_2}, \quad (26)$$

where σ controls how quickly affinity decays with prototype distance. At inference, class scores are $\log a_{ic}$ followed by softmax normalization. The affinities $a_{ic} \in (0, 1]$ are geometric similarity scores, not class probabilities: they are trained with a margin objective and never optimized to satisfy likelihood or calibration constraints. Applying softmax to $\log a_{ic}$ therefore yields a convenient ranking score for argmax decisions, but it should not be read as a calibrated estimate of $P(y = c \mid \mathbf{x})$ without additional post-hoc mapping.

Training optimizes a margin term on affinities rather than cross-entropy on logits. For batch index i with label y_i , define the per-example margin loss

$$\ell_i = \sum_{c \neq y_i} \max(0, \lambda + a_{ic} - a_{i,y_i}), \quad (27)$$

which penalizes wrong-class affinities that exceed the true-class affinity by less than margin λ . A diversity regularizer $R(\mathbf{W})$ encourages prototypes to spread apart by penalizing deviations of pairwise squared distances from their mean. The total objective combines class reweighting, a center-focused local weight, and prototype diversity:

$$\mathcal{L}_{\text{CFAL}} = \frac{1}{B} \sum_{i=1}^B \frac{1}{E_{y_i}} (1 - a_{i,y_i})^\gamma \ell_i + R(\mathbf{W}), \quad (28)$$

where E_c is the effective number of training examples in class c derived from class count n_c and hyperparameter β , and γ up-weights examples whose true-class affinity remains low. Unlike the CE plus soft-F1 and soft-MCC hybrids, CFAL does not use balanced oversampling: it is categorized as an algorithm-level patch method because it changes the head and loss while keeping the natural training distribution.

Thesis variant. The source defaults $\lambda = 0.1$, $\beta = 0.999$, and $\gamma = 2.0$ are held fixed. Only the affinity bandwidth σ is selected on validation. It controls the distance scale in Equation (26), not the strength of rebalancing, and no value recovers ordinary CE.

The published $\sigma = 130$ was calibrated for unnormalized AlexNet activations. Here, both representations and prototypes are L2-normalized, which bounds their squared distance to $[0, 4]$. At $\sigma = 130$ all affinities would therefore exceed 0.969, leaving too little contrast for the margin loss. The tested range is instead matched to the normalized geometry. Normalization also prevents uncontrolled feature magnitude from changing the affinity scale across training conditions.

Because normalization and validation-selected σ depart from the source, the method is reported as *CFAL-inspired*. The margin loss, effective-number weight, affinity difficulty factor, and prototype diversity regularizer otherwise follow the source. Evaluation is limited to patch classification because CFAL requires instance-level labels rather than weak slide-level bags.

3.4 Hybrid Sampling–Loss Methods

Hybrid methods combine data-level and algorithm-level changes. The Scholz-style patch objectives below adapt the metric-derived losses of Scholz et al. for imbalanced medical classification (Scholz et al., 2024). Both tested variants use class-balanced oversampling together with metric-derived losses, so their effect cannot be attributed only to a new loss or only to a new sampler.

3.4.1 Scholz-Style CE + Soft F1

The intuition is that CE optimizes the log-probability of the correct class one example at a time, whereas macro F1 evaluates a different object: the balance between precision and recall at class level. Under class imbalance, this distinction matters because a model can reduce CE substantially by fitting frequent classes while still leaving tail-class recall or precision weak.

The obstacle is that ordinary F1 is computed from hard counts after prediction, so it is not directly usable as a gradient-based training loss. The Scholz-style soft-F1 objective replaces those hard counts with probability-weighted counts inside each minibatch. For one class, a confident correct prediction contributes almost one soft true positive, a confident prediction for the wrong class contributes to soft false positives, and low probability on a true class contributes to soft false negatives. Formally, for one-hot labels y_i and predicted probabilities $\hat{y}_i$:

$$\text{TP} = \sum_i y_i \hat{y}_i, \quad \text{FP} = \sum_i (1 - y_i) \hat{y}_i, \quad \text{FN} = \sum_i y_i (1 - \hat{y}_i). \quad (29)$$

These terms keep the familiar meaning of a confusion matrix but make every component differentiable with respect to the model probabilities. They then define soft precision, soft recall, and soft F1:

$$\text{F1}_{\text{soft}} = \frac{2 \text{precision}_{\text{soft}} \text{recall}_{\text{soft}}}{\text{precision}_{\text{soft}} + \text{recall}_{\text{soft}}}. \quad (30)$$

The multiclass implementation applies this construction separately to each class and averages the resulting classwise terms. This macro-style averaging is the imbalance-aware part: each class contributes one term regardless of how many samples it has in the full dataset. The tested objective does not replace CE completely; it adds the metric-derived term to CE so that the objective retains a likelihood term while also pushing the minibatch toward higher macro soft F1:

$$\mathcal{L}_{\text{CE+F1}} = \mathcal{L}_{\text{CE}} + \lambda_{\text{aux}} \left(1 - \frac{1}{C} \sum_{c=1}^C \text{F1}_{\text{soft},c} \right). \quad (31)$$

The auxiliary weight $\lambda_{\text{aux}} > 0$ controls the contribution of the soft-F1 term and is selected on validation. This loss is trained with balanced sampling on the patch training set. That design is important for interpretation: the loss tries to optimize a macro-style class metric, while the sampler ensures that tail classes occur often enough in minibatches for their soft-F1 terms to provide a regular training signal. The retained CE term should not be read as preserving proper-scoring-rule behaviour. CE alone is minimized by the true conditional probabilities, but the soft-F1 term is not, and their sum is therefore no longer a proper scoring rule: its minimizer trades probability accuracy against the metric at a rate set by λ_{aux} . The same holds for the soft-MCC hybrid below. Both are consequently reported with calibration measured rather than assumed.

3.4.2 Scholz-Style CE + Soft MCC

The soft-MCC variant uses the same principle as soft F1—replace hard evaluation counts with differentiable probability masses—but starts from the Matthews correlation coefficient (MCC). Whereas F1 for one class depends mainly on that class’s positive predictions, MCC is built from the full confusion matrix and therefore reacts to false positives, false negatives, and true negatives at the same time. In binary problems it is often preferred in imbalanced settings because it can remain informative when one class dominates the dataset: accuracy can look high while predictions are still systematically biased, but MCC penalizes both kinds of mistakes jointly.

Binary MCC as a correlation coefficient. For one class treated as positive, write TP, TN, FP, and FN for the four confusion-matrix cells. The standard MCC is

$$\text{MCC} = \frac{\text{TP} \cdot \text{TN} - \text{FP} \cdot \text{FN}}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}}. \quad (32)$$

The numerator compares co-occurrence of correct positive and correct negative decisions with co-occurrence of the two error types; the denominator scales by how many examples fall into

each margin of the confusion matrix. MCC equals +1 for perfect predictions, 0 for chance-level association, and can become negative when predictions are worse than random. This makes it a correlation-style score rather than a prevalence-weighted rate.

Multiclass MCC in a minibatch. For C target classes, the multiclass extension treats labels and predictions as C -dimensional vectors per example. Let $Y \in \{0, 1\}^{N \times C}$ be the one-hot label matrix and $\hat{Y} \in [0, 1]^{N \times C}$ the corresponding softmax probabilities, with rows $\mathbf{y}_i$ and $\hat{\mathbf{y}}_i$. Three aggregate quantities describe how mass is distributed inside the batch:

$$s = \sum_{i=1}^N \sum_{c=1}^C Y_{ic} \hat{Y}_{ic}, \quad t_c = \sum_{i=1}^N Y_{ic}, \quad p_c = \sum_{i=1}^N \hat{Y}_{ic}. \quad (33)$$

Here s is the total probability-weighted agreement between labels and predictions: each example contributes $\hat{y}_{i, y_i}$, the predicted mass on its true class. The terms t_c and p_c are the batch totals of true and predicted mass for class c . They play the role of marginals in the confusion-matrix calculation. If predictions collapse onto a few head classes, the vector $(p_1, \dots, p_C)$ becomes skewed even when argmax accuracy looks acceptable; MCC is sensitive to that mismatch because it compares the joint alignment s with what would be expected from the marginals alone.

The multiclass MCC used in training can be written as a normalized covariance between the label matrix and the prediction matrix:

$$\text{MCC}_{\text{mc}} = \frac{Ns - \sum_{c=1}^C t_c p_c}{\sqrt{N^2 - \sum_{c=1}^C p_c^2} \sqrt{N^2 - \sum_{c=1}^C t_c^2}}. \quad (34)$$

The numerator $Ns - \sum_c t_c p_c$ increases when true and predicted mass line up example by example and decreases when correct predictions are no better than what the marginal totals would suggest by chance. The two square-root terms measure how spread out the batch-level label totals and prediction totals are across classes. They do not, however, make few-class batches harmless. A minibatch whose examples all carry the same label has $t_c = N$ for that class and zero elsewhere, so $N^2 - \sum_c t_c^2 = 0$ and the denominator vanishes; the prediction-side term degenerates in the same way when the softmax outputs collapse onto one class. MCC is undefined in exactly those batches rather than merely large, which is a case the implementation has to handle explicitly. Equation (34) reduces to the familiar binary form in (32) when $C = 2$.

Soft MCC loss. Hard MCC replaces $\hat{Y}$ by one-hot argmax predictions, which again blocks gradient-based training. The soft-MCC objective keeps the structure of (34) but plugs in softmax probabilities directly:

$$\text{MCC}_{\text{soft}} = \frac{Ns - \sum_{c=1}^C t_c p_c}{\sqrt{N^2 - \sum_{c=1}^C p_c^2} \sqrt{N^2 - \sum_{c=1}^C t_c^2}}. \quad (35)$$

Every term in (33) and (35) is differentiable with respect to the logits that produce $\hat{Y}$. A confident correct prediction increases s ; probability placed on the wrong classes increases some p_c while leaving the matching t_c unchanged, which lowers the numerator; and overly peaked prediction vectors reduce the prediction-side denominator term, limiting the score even if a few examples are classified correctly. The degenerate batches identified above are handled numerically rather than left to the algebra: both spread terms are floored at $\epsilon_{\text{spr}} = 1 \times 10^{-8}$ before the square roots are taken, and a further $\epsilon = 1 \times 10^{-8}$ is added to the resulting denominator, so a single-class batch contributes a bounded term with a vanishing gradient instead of a division by zero. Soft F1 is stabilized the same way, with $\epsilon_{\text{F1}} = 1 \times 10^{-16}$ added to the denominator

$2\text{TP} + \text{FP} + \text{FN}$ of Equation (30), which is zero for a class absent from the batch and unpredicted in it; that class then contributes a full unit of loss, which is the intended macro-averaging behaviour. Because balanced sampling supplies the minibatches for both hybrids, single-class batches are rare at the batch sizes used, but the floors are part of the specification rather than a fallback. With that in place, the training objective turns a score to be maximized into a loss to be minimized and adds it to CE:

$$\mathcal{L}_{\text{CE}+\text{MCC}} = \mathcal{L}_{\text{CE}} + \lambda_{\text{aux}} (1 - \text{MCC}_{\text{soft}}). \quad (36)$$

The auxiliary weight $\lambda_{\text{aux}} > 0$ has the same role and validation grid as in the soft-F1 hybrid. This variant is also evaluated only in the patch regime and is trained with class-balanced over-sampling. The soft-F1 and soft-MCC variants therefore ask a focused question: once tail classes are regularly presented to the optimizer, does adding a differentiable version of a class-balanced evaluation metric improve the controlled patch classifier beyond CE and simple balanced sampling?

3.4.3 Odd-K-Out (OKO) Set Learning

Standard cross-entropy optimizes each example independently, which has two relevant consequences under class imbalance. First, the gradient signal for a tail class arrives only when that class is sampled, so rare examples can be overwhelmed by frequent ones. Second, hard-label cross-entropy encourages logit values to diverge without bound, which is a known driver of overconfidence and poor calibration (Guo et al., 2017). OKO (Muttenthaler et al., 2024) addresses both problems simultaneously by replacing per-example cross-entropy with a set-level objective. Calibration is inherently a property of a model’s output distribution over a collection of examples, and training on sets is a natural way to expose the model to that distributional structure during learning.

Set construction. A training set S of size $k + 2$ is constructed as follows. A pair class y' is sampled uniformly from $[C]$, equalizing class frequency in expectation regardless of the training distribution—analogous to balanced sampling but applied at the set level rather than the sample level. Two examples x'_1, x'_2 are drawn from the class y' , and k distinct odd classes $y'_3, \dots, y'_{k+2}$ are sampled without replacement from $[C] \setminus \{y'\}$. One representative example x'_j is drawn from each odd class y'_j . The resulting set $S = (x'_1, \dots, x'_{k+2})$ therefore always contains exactly one matched pair and k distractors from different classes.

Set-aggregated loss. For a model f_θ parameterized by θ , OKO defines the aggregated logit vector as the sum of individual per-example outputs over the set members:

$$f_\theta(S) = \sum_{i=1}^{k+2} f_\theta(x'_i). \quad (37)$$

Prediction on the aggregated logits uses ordinary softmax. The pair class y' appears twice in the set and its two logit contributions therefore reinforce each other, raising the aggregate score for y' relative to any single odd class. The aggregation is a sum, not an average: adding members raises the magnitude of $f_\theta(S)$ rather than damping it, so the calibration behaviour cannot be attributed to pooling shrinking the logits. It arises instead from the objective the sum is placed inside. Predicting the pair class from an aggregate that also carries k odd-class contributions means the gradient rewards a member’s logits only insofar as they help the set-level decision, and Muttenthaler et al. (2024) derive from this an implicit entropic regularization of the learned representation—a bias toward less extreme, higher-entropy outputs—which is what they link to improved calibration under hard labels.

The main training loss predicts the pair class from the aggregated logits:

$$\ell_{\text{OKO}} = -\log \text{softmax}(f_{\theta}(S))_{y'}. \quad (38)$$

Following the main-text configuration of the original paper, the implementation also trains an auxiliary classification head g_{θ} that shares the trunk with f_{θ} and predicts the first odd class y'_3 :

$$\ell_{\text{aux}} = -\log \text{softmax}(g_{\theta}(S))_{y'_3}. \quad (39)$$

The total per-set loss is $\ell_{\text{OKO}} + \ell_{\text{aux}}$. The auxiliary head is discarded at inference, and single-example evaluation proceeds exactly as for any other classifier: the model accepts one patch feature vector and returns the logits from the main head. There is no inference overhead or architecture change at test time.

OKO belongs in the hybrid sampling–loss category because the set construction couples a balanced class-sampling policy with a modified training objective; neither component is meaningful on its own. Balanced sampling via uniform pair-class selection ensures that all C classes contribute equally to the gradient signal in expectation, which directly addresses the data-level imbalance problem. The aggregated set-level loss then introduces an implicit regularization on logit magnitude. Together these are not separable in the same way that standard cross-entropy and a `WeightedRandomSampler` are separable. OKO is evaluated in the patch feature regime only; the set-construction mechanism does not extend naturally to the attention-MIL bag encoder of the WSI-bag regime.

The control k is the number of odd classes in a set, and therefore the set cardinality $k + 2$; it is the tuned control selected on validation. It fixes how many distinct classes a single aggregate must discriminate among, not a degree of averaging and not a rebalancing strength. The paper’s ablation experiments show that accuracy and calibration are insensitive to the specific value of k , and its main configuration uses $k = 1$ together with the auxiliary odd-class head of Equation (39); $k = 1$ is also the computationally cheapest option, though a moderate set size can work best in practice without requiring the largest sets. Because a set consumes $k + 2$ forward passes, the comparison against CE is made at a matched number of optimizer updates, as in the source, rather than at matched examples seen.

3.5 Representation and Architecture-Level Methods

Representation-level methods try to make rare classes more separable in feature space rather than only changing sampling or per-example loss weights. This family is especially relevant for computational pathology, where WSI labels supervise bags of many unlabeled instances and class imbalance can occur both across slides and within a positive slide (Juyal et al., 2024).

3.5.1 Classifier Re-Training (cRT)

Classifier re-training (cRT) separates representation learning from classifier fitting (Kang et al., 2020). Stage one trains the ordinary CE model on the observed imbalanced training distribution. Stage two freezes the learned representation, discards and reinitializes the final linear classifier, and fits only that classifier using class-balanced sampling. If $h_{\hat{\theta}}(x)$ is the frozen representation learned in stage one, cRT estimates

$$(\widehat{W}, \widehat{\mathbf{b}}) = \arg \min_{W, \mathbf{b}} \mathbb{E}_{(x, y) \sim q_{\text{bal}}} \left[-\log \text{softmax}(Wh_{\hat{\theta}}(x) + \mathbf{b})_y \right], \quad (40)$$

where q_{bal} gives classes equal expected exposure. This isolates whether imbalance primarily distorts the decision boundary after a useful representation has already been learned.

In the patch regime, stage one learns the common MLP head on frozen Virchow2 embeddings; cRT freezes its learned hidden representation and retraining a reinitialized linear output

layer with balanced patch sampling. In the WSI regime, stage one learns the instance projection and attention aggregator; cRT freezes both components and retrain only a reinitialized bag classifier with balanced slide sampling. Validation selects the stage-two checkpoint, and its compute is reported in addition to stage-one training. The original cRT study considers several classifier normalization variants; the tested implementation uses only the reinitialized linear-classifier variant specified here.

3.5.2 SC-MIL

SC-MIL extends supervised contrastive learning to MIL by applying the contrastive objective to bag representations instead of assigning noisy slide labels to individual patches (Juyal et al., 2024). The motivation is that a slide label usually does not describe every patch in a bag. A cancer-type label can be correct for the slide while many instances contain background, normal tissue, technical noise, or weakly informative morphology. Applying contrastive learning directly to instances would therefore treat many patches as positives or negatives using labels they do not truly carry. SC-MIL avoids that mismatch by contrasting the aggregated slide-level representation.

The attention-MIL backbone first maps bag B_i to an embedding $\mathbf{b}_i = \mathbf{b}(B_i)$. A projection head g then produces normalized contrastive features $\mathbf{r}_i = g(\mathbf{b}_i)$. In the contrastive view, each bag in a minibatch is an anchor. Bags with the same cancer-type label are positives because their slide-level representations should become closer, while bags from other classes act as negatives because their representations should remain separated. For anchor i , bags of the same class form the positive set $P(i)$, and the supervised contrastive term is

$$\mathcal{L}_{\text{SCL}} = -\frac{1}{\sum_i |P(i)|} \sum_i \sum_{p \in P(i)} \log \frac{\exp(\mathbf{r}_i^\top \mathbf{r}_p / \tau)}{\sum_{a \neq i} \exp(\mathbf{r}_i^\top \mathbf{r}_a / \tau)}. \quad (41)$$

The numerator rewards similarity to same-class bags. The denominator compares that similarity against all other bags in the minibatch, so the loss favors representations where class clusters are compact and mutually separated. The temperature τ controls how sharply these similarities are weighted. The source reference is $\tau = 1.0$; it is retained in the validation grid together with lower temperatures.

SC-MIL does not rely on contrastive learning alone. The contrastive term shapes the representation space, while CE trains the final class predictor. The tested variant combines them through a linear curriculum:

$$\mathcal{L}_t = \beta_t \mathcal{L}_{\text{SCL}} + (1 - \beta_t) \mathcal{L}_{\text{CE}}, \quad \beta_t = 1 - \text{progress}_t. \quad (42)$$

Early training therefore emphasizes class-structured bag embeddings, when the classifier head is still unstable. As training progresses, β_t decreases and more weight moves to CE, so the learned representation is tied back to the actual slide-level classification objective. Class-aware batching draws at least two slides for each represented class whenever class support permits; otherwise an anchor would have $|P(i)| = 0$ and contribute no contrastive term. Training logs the number of valid anchors, ordered positive pairs $\sum_i |P(i)|$, and represented classes per batch. The normalization in Equation (41) uses the actual number of ordered positive pairs, so unequal class multiplicities do not silently change the scale of the contrastive objective.

3.5.3 MDE-Inspired Dual-Expert MIL

The MDE-inspired variant uses distribution-specific experts and a cross-expert consistency term for long-tailed WSI classification (Ling et al., 2025). The full source method combines a dual-expert ensemble with multimodal distillation from a pathology foundation model. The tested implementation includes only the ensemble component: a shared attention aggregator, two

expert classification heads, and optional logit consistency under the same frozen Virchow2 bags as the other MIL methods. It omits CONCH-based multimodal distillation and is therefore not called a faithful MDE-MIL reproduction.

The shared trunk matches plain attention MIL: each bag $B = \{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ is encoded instance-wise, attention weights are computed with a linear scorer and softmax over instances, and the weighted sum yields a bag embedding $S(B) \in \mathbb{R}^d$. Two separate feed-forward experts then map S to logits:

$$E_U(S) = W_U \phi_U(S) + b_U, \quad E_B(S) = W_B \phi_B(S) + b_B, \quad (43)$$

where ϕ_U and ϕ_B are two-layer MLPs with ReLU and dropout. Expert U is trained on the natural slide distribution and expert B on a class-balanced slide distribution.

Each training step draws one minibatch from each distribution. The unbalanced loader shuffles slide bags as in MIL CE; the balanced loader uses the same inverse-frequency weights as balanced-sampling MIL. For batches (B_U, y_U) and (B_B, y_B) , the classification loss is

$$\mathcal{L}_{\text{cls}} = \text{CE}(E_U(S(B_U)), y_U) + \text{CE}(E_B(S(B_B)), y_B), \quad (44)$$

and the cross-expert consistency term penalizes disagreement on the *same* bag embedding:

$$\mathcal{L}_{\text{con}} = \text{MSE}(E_U(S(B_U)), E_B(S(B_U))) + \text{MSE}(E_B(S(B_B)), E_U(S(B_B))). \quad (45)$$

The total objective is $\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{con}} \mathcal{L}_{\text{con}}$. The reference default is $\lambda_{\text{con}} = 0.25$, matching the Camelyon+-LT default reported by Ling et al.; λ_{con} is selected on validation. Setting $\lambda_{\text{con}} = 0$ defines the prespecified *dual experts without consistency* component ablation: both natural and balanced experts remain, but only their classification losses are optimized. Both experts share the aggregator throughout training.

At validation and test time, each slide bag is evaluated once in natural order with mean-logit ensemble inference:

$$\hat{\mathbf{z}}(B) = \frac{1}{2}(E_U(S(B)) + E_B(S(B))). \quad (46)$$

MDE-inspired dual-expert MIL is categorized as a hybrid WSI method because it couples two sampling distributions with an optional consistency regularizer rather than changing only the sampler or only the per-example loss.

4 Summary

This report places the tested roster on two orthogonal axes—the signal a method consults to infer that a class or example needs help (prevalence, support in its nominal and independent readings, difficulty, diversity) and the level at which it intervenes (sampling, objective, prior correction, representation)—and defines every method in the input form it consumes, with CE and MIL CE as the no-mitigation references (Table 2). Most collected methods read prevalence; the remaining signals are covered by class-balanced CE, pilot-difficulty weighted CE, semantic-scale weighted CE, and independent-support weighted CE, a thesis construction. Five of them share one intervention level—a mean-one class weight on unmodified CE—which is what makes the signal comparison interpretable.

Where a source method could not be reproduced as published, the deviation is stated with the method and the result labelled accordingly. Output semantics are recorded where an intervention changes them, prior-corrected and temperature-scaled scores being kept separate from raw ones. These definitions are what the evaluation of discrimination, probability quality, and mitigation recovery presupposes.

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